

## Homework 3

1) 7 door Monty Hall

- initially all possibilities are equally likely
- Monty opens 3 doors
- You can switch to 3 remaining doors

2) P(success switching)

$C_i$  : event car is behind door  $i$

$W$  : success switching

WLOG assume contestant picked door 1

$$\begin{aligned} P(W) &= P(W|C_1)P(C_1) + P(W|C_2)P(C_2) \\ &\quad + P(W|C_3)P(C_3) + P(W|C_4)P(C_4) \\ &\quad + P(W|C_5)P(C_5) + P(W|C_6)P(C_6) \\ &\quad + P(W|C_7)P(C_7) \\ &= 6 \cdot \frac{1}{3} \cdot \frac{1}{7} = \frac{2}{7} \end{aligned}$$

b) Generalized:  $n \geq 3$  doors

Monty opens  $m$  goat doors  $1 \leq m \leq n-2$

$$P(W) = (n-1) \frac{1}{n-1-m} \cdot \frac{1}{n} = \frac{n-1}{(n-m-1)n}$$

2) odds of event w/ prob  $p$  defined by  $\frac{p}{1-p}$

hypothesis  $H$  (event  $H$ )

gather new data/evidence (event  $D$ )

prior — prob for  $H$  being true before gathering new data

posterior — " " " " " AFTER " " "

likelihood ratio:  $\frac{P(D|H)}{P(D|\neg H)}$

Answer key just takes  $\frac{P(H|D)}{P(\neg H|D)}$

$$\frac{P(H|D)}{P(H^c|D)} = \frac{P(H)}{P(H^c)} \frac{P(D|H)}{P(D|H^c)}$$

...  
2nd does Bayes  
on numerator &  
denominator...

$$\frac{\cancel{P(D|H)} P(H)}{P(D) P(H^c|D)} = \frac{\cancel{P(H)} P(D|H)}{P(H^c) P(D|H^c)}$$

$$P(D) P(H^c|D) = P(H^c) P(D|H^c)$$

$$P(D) (1 - P(H|D)) = P(H^c) P(D|H^c)$$

$$P(D) = P(H|D) P(D) + P(D|H^c) P(H^c)$$

$$= P(D|H) P(H) + P(D|H^c) P(H^c)$$

LOOP!

- b) D: has disease  
A: first test positive  
S: 2nd " "

2 steps

$$P_n(D) = P(D|F) = \frac{P(D)}{P(D^c)} P(D^c|F) \frac{P(F|D)}{P(F|D^c)}$$

$$= \frac{1}{99} \times 0.95 \times \frac{P(D^c|F)}{P(F|D^c)}$$

$$= \frac{1}{99} \times 0.95 \times \frac{\cancel{P(F|D^c)} P(D^c)}{\cancel{P(F|D^c)} P(F|D^c)}$$

$$= \frac{0.99}{99} \times 0.95 \times \frac{1}{P(F)}$$

$$= 0.0095 \times \frac{1}{P(F|D)P(D) + P(F|D^c)P(D^c)}$$

$$= 0.0095 \times \frac{1}{0.95 \times 0.01 + 0.05 \times 0.99}$$

$$= \frac{95}{890}$$

$$P_n(D|S) = \frac{P_n(D)}{P_n(D^c)} P_n(D^c|S) \frac{P_n(S|D)}{P_n(S|D^c)}$$

$$= \frac{95}{495} \times 0.95 \times \frac{P_n(D^c|S)}{P_n(S|D^c)}$$

$$\begin{aligned}
 &= \frac{95}{495} \times 0.95 \times \frac{P_n(CD^c)}{P_n(CS)} \frac{495}{190} \\
 &= \frac{95}{890} \times 0.95 \times \frac{1}{P_n(CSID)P_n(CD) + P_n(CSID^c)P_n(CD^c)} \\
 &= \frac{361}{460}
 \end{aligned}$$

1 step

$$\frac{P(D|S, F)}{P(D^c|S, F)} = \frac{P(D|F)}{P(D^c|F)} \frac{P(CSID, F)}{P(CSID^c, F)}$$

$$\begin{aligned}
 P(D|S, F) &= \frac{P(D|F)}{P(D^c|F)} \frac{P(D^c|S, F)}{P(CSID^c)} \frac{P(CSID)}{P(CSID^c)} \\
 &= \frac{P(D|F)}{P(D^c|F)} \frac{\cancel{P(CSID, F)} P(D^c|F)}{P(CSID, F)} \frac{P(CSID)}{P(CSID^c)} \\
 &= \frac{P(D|F)}{P(CSID, F)} \frac{P(CSID)}{P(CSID^c)} \\
 &= \frac{P(CSID, F)}{P(CSID, F)P(D|F) + P(CSID^c, F)P(D^c|F)} \\
 &= \frac{361}{460}
 \end{aligned}$$

Answer key calculates odds using the equation  
and converts odds to probabilities!

$$\frac{1}{99} \cdot \frac{0.95^2}{0.05^2} = \frac{361}{99} \rightarrow \frac{361}{460}$$

$$\text{1st update: } \frac{1}{99} \frac{0.95}{0.05}$$

these are prior odds for...

$$\text{2nd update: } \left( \frac{1}{99} \frac{0.95}{0.05} \right) \frac{0.95}{0.05}$$

3)  $A_1, A_2, B, C$

$$P(A_1|B) > P(A_1|C) \quad P(A_2|B) > P(A_2|C)$$

$$P(A_1 \cup A_2|B) < P(A_1 \cup A_2|C)$$

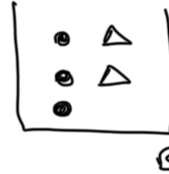
Yes, this is possible.

$A_1$ : draw red pebble

$A_2$ : draw dark pebble

$B$ : drawing from basket B

$C$ : drawing from basket C



4) a)  $P(C \text{ wins an independent game}) = p$

$C$ : event  $C$  wins entire game

$$P(C) = P(C| \text{wins first game}) P(\text{wins first game}) \\ + P(C| \text{loses first game}) P(\text{loses first game})$$

$$= P(C| \text{wins first}) p + P(C| \text{loses first}) q$$

$$= p [ P(C| \text{wins second}) P(\text{wins second}) \\ + P(C| \text{loses second}) P(\text{loses second}) ]$$

$$+ q [ P(C| \text{loses second}) P(\text{loses second}) \\ + P(C| \text{wins second}) P(\text{wins second}) ]$$

$$P(C) = p^2 + 2pq P(C)$$

$$P(C) = \frac{p^2}{1-2pq}$$

$$b) \quad p = \frac{1 - (q/p)^2}{1 - (q/p)^4} = \frac{p^2 - q^2}{p^4 - q^4} \quad p^2 = \frac{p^2}{p^2 + q^2} = \frac{p^2}{1 - 2pq}$$

5) a)  $p_i$  is probability of  $S_i$

$V_i$  = value of dice roll  $i$   $i \geq 1$

$$S_i = V_1 + \dots + V_i$$

$$\begin{aligned} p_n &= p_{n-6} P(\text{roll } 6) + p_{n-5} P(\text{roll } 5) \\ &\quad + p_{n-4} P(\text{roll } 4) + p_{n-3} P(\text{roll } 3) \\ &\quad + p_{n-2} P(\text{roll } 2) + p_{n-1} P(\text{roll } 1) \\ &= \frac{1}{6} (p_{n-1} + p_{n-2} + \dots + p_{n-6}) \end{aligned}$$

b)  $p_1 = p_0 r_1 = \frac{1}{6}$

$$p_2 = p_0 r_2 + p_1 r_1 = \frac{7}{36}$$

$$p_3 = p_0 r_3 + p_1 r_2 + p_2 r_1 = \frac{49}{216}$$

$$p_4 = p_0 r_4 + p_1 r_3 + p_2 r_2 + p_3 r_1 = \frac{343}{6^4}$$

$$p_5 = \dots \frac{7^4}{6^5}$$

$$p_6 = \dots \frac{7^5}{6^6}$$

$$p_7 = \frac{1}{6} (p_1 + \dots + p_6) = \frac{70993}{279336} \approx 0.2593$$

c) Stumped.

Answer key: average value thrown is  $\frac{7}{2}$

therefore we can expect to  
land every 3.5 numbers

6) a)  $A \sim \text{Bin}(n, p_1)$

PMF is  $\binom{n}{k} p_1^k q_1^{n-k}$   $k \in \{0: n\}$

b)

$$P(A+B=i) = \sum_{j=0}^i \binom{m}{j} p_1^j q_1^{m-j} \binom{n}{i-j} p_2^{i-j} q_2^{n-i+j}$$

↳ becomes  $\text{Bin}(n+m, p)$  when i.i.d  
 $p_1 = p_2 = p$

c)  $P(A \text{ wins}) = p_1 + q_1 q_2 p_1 + (q_1 q_2)^2 p_1 + \dots$

$$= p_1 + q_1 q_2 P(A \text{ wins})$$

$$P(A \text{ wins}) = \frac{p_1}{1 - q_1 q_2}$$

Answer key conditions on first question  
 for each player

$$\begin{aligned} P(A \text{ wins}) &= p_1 + (1-p_1)p_2 \cdot 0 + (1-p_1)(1-p_2) P(A \text{ wins}) \\ &= \frac{p_1}{1 - q_1 q_2} \end{aligned}$$

7)2) As long as an even # of bits are corrupted,  
 no error detected

$$\binom{5}{2} (0.1)^2 (0.9)^3 + \binom{5}{4} (0.1)^4 (0.9) = 0.07835$$

b)  $\sum_{\substack{i \text{ is even} \\ i \geq 2}} \binom{n}{i} p^i (1-p)^{n-i}$

c)  $a+b = \sum_{k=0}^n \binom{n}{k} p^k (1-p)^{n-k} = 1$

→ ... → n-k

$$z-b = \sum_{k=0}^n \binom{n}{k} (1-p)^k (1-p)^{n-k} = (1-2p)^n$$

$$a = \frac{1 + (1-2p)^n}{2}$$

$$P(\text{no errors detected} | \text{error}) = \frac{1 + (1-2p)^n}{2} - (1-p)^n$$